

EDGAR Financial Engine — Taxonomy Research Report

Morningstar / Rule One Toolbox / EDGAR XBRL Cross-Source Validation & Gap Analysis

Date: 2026-03-09 **Scope:** Validate and expand the `edgarFinancials.js` XBRL taxonomy to fully support Morningstar-quality financial statements and all Rule One Toolbox calculations/metrics.

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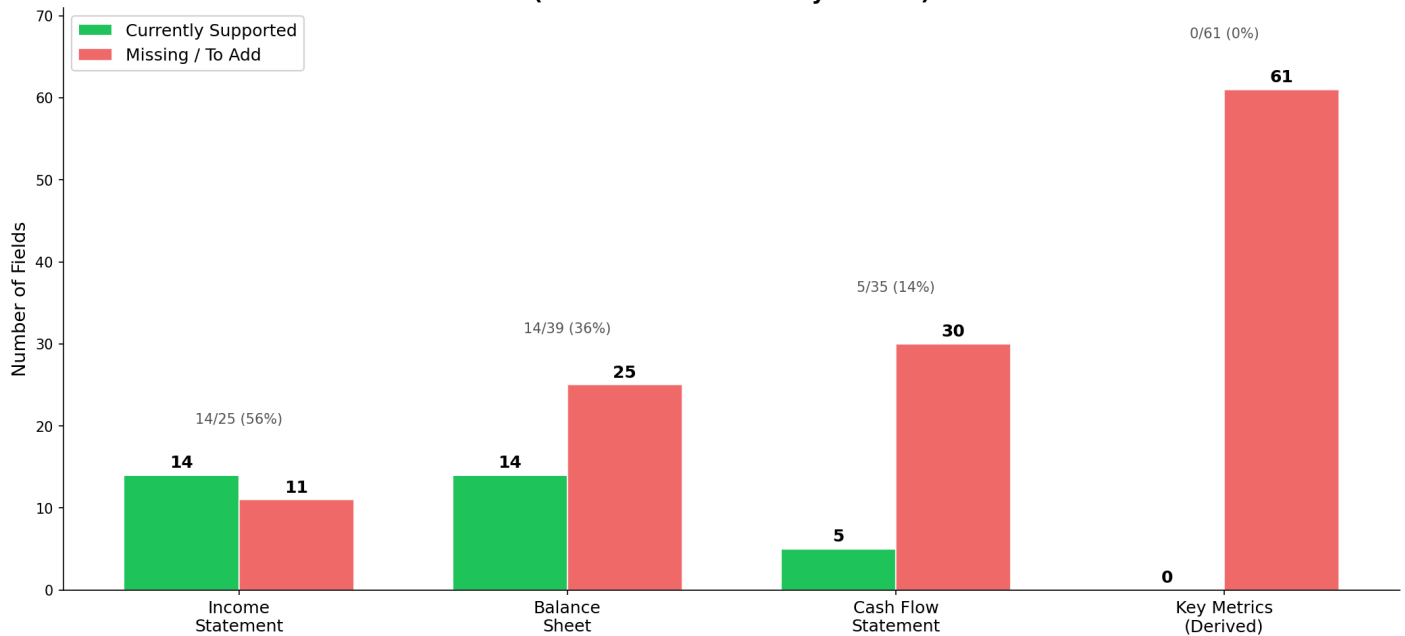
1. Executive Summary

Key Findings

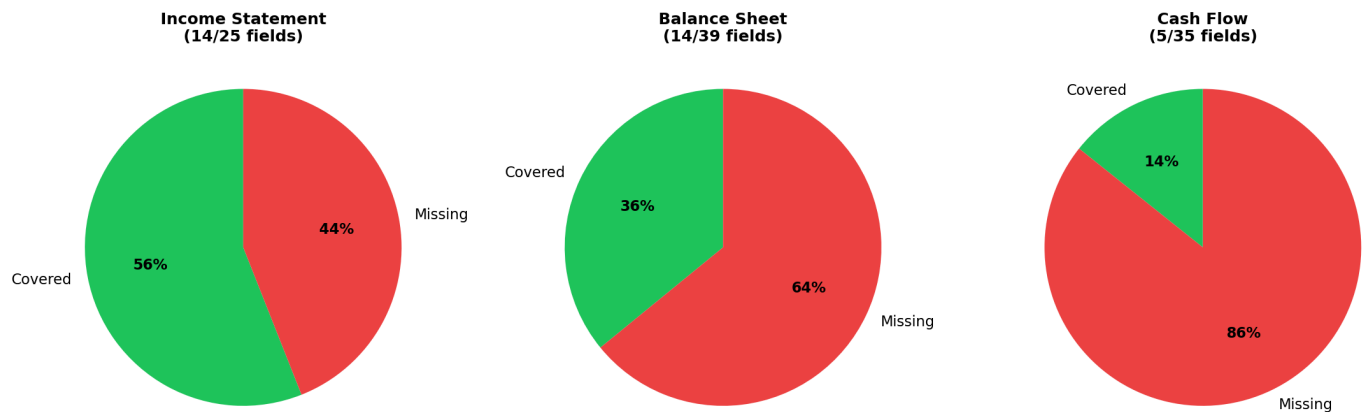
- **Current engine coverage:** 33 of 99 Rule One Toolbox consolidated financial fields (33%) are supported in `edgarFinancials.js`
- **Income Statement:** 14/25 fields covered (56%)
- **Balance Sheet:** 14/39 fields covered (36%)
- **Cash Flow:** 5/35 fields covered (14%)
- **Key Metrics:** 0/61 derived metrics currently computed (these depend on the missing statement fields)
- **Cross-source validation:** Morningstar and Rule One Toolbox report **identical values** for all core financial items — they both normalize from the same SEC EDGAR source. The only differences are sign conventions and unit scaling.
- **EDGAR has the data:** Every line item in both Morningstar and Rule One Toolbox traces back to SEC EDGAR XBRL tags. Nothing requires a paid API.

Coverage Visualization

edgarFinancials.js Coverage: Current vs Required Fields (Rule One Toolbox + Key Metrics)



Current edgarFinancials.js Coverage vs Rule One Toolbox Requirements



What Needs to Happen

The taxonomy in `edgarFinancials.js` needs to expand from ~36 XBRL tag mappings to ~80+. Additionally, ~25 derived/calculated fields need to be added (e.g., Gross PP&E, Total Debt, Net Debt, EBIT, EBITDA, working capital changes, liquidity ratios). With these additions, the engine will support:

- Full Morningstar-style expandable financial statements
- All Rule One Toolbox calculations and metrics
- Complete Rule One Score computation without manual inputs

2. Data Sources & Methodology

Sources Analyzed

Source	Files	Format	Companies	Periods
Morningstar — As Originally Reported	6 XLS	Hierarchical with indent levels	AAPL, SFM	2021-2025 + TTM
Morningstar — Restated	6 XLS	Hierarchical with indent levels	AAPL, SFM	2021-2025 + TTM
Rule One Toolbox — Consolidated Financials	2 CSV	Flat, 3 statement sections	AAPL, SFM	2016-2025 + TTM
Rule One Toolbox — Key Metrics	2 CSV	Flat, 6 metric categories	AAPL, SFM	Up to 47 years (AAPL)
SEC EDGAR XBRL	API	JSON, tag-based facts	AAPL, SFM	All available years
Current Engine (edgarFinancials.js)	1 JS	3 taxonomy arrays	All companies	All EDGAR years

Methodology

- 1. Parsed all 12 Morningstar XLS files using Python `xlrd` — extracted field names, hierarchy levels, and values
- 2. Parsed all 4 Rule One Toolbox CSVs — extracted field names, sections, and values
- 3. Read the current `edgarFinancials.js` taxonomy (3 arrays: INCOME, BALANCE, CASHFLOW)
- 4. Cross-validated actual numbers for SFM and AAPL (FY2025) across all sources
- 5. Fetched live EDGAR XBRL data to confirm tag availability and values
- 6. Mapped every line item to its EDGAR XBRL tag(s) with fallback chains

Unit Conventions

Source	Financial Amounts	Per-Share Data	Share Counts
Morningstar	Raw dollars (e.g., 8806159000)	Dollars (e.g., 5.31)	Raw shares (e.g., 97687000)
Rule One Toolbox	Millions (e.g., 8806.159)	Dollars (e.g., 5.31)	Millions (e.g., 97.687)
EDGAR XBRL	Raw dollars	Dollars	Raw shares
Current Engine	Raw dollars (follows EDGAR)	Dollars	Raw shares

Sign Conventions

Source	Expenses	CapEx	Buybacks	Dividends
Morningstar	Negative (e.g., -5389770000)	Negative	Negative	Negative
Rule One Toolbox	Positive (e.g., 5389.77)	Negative	Negative	Negative
EDGAR XBRL	Positive (expenses are positive amounts)	Positive (payments are positive)	Positive	Positive

3. Morningstar Financial Statement Taxonomy

Morningstar provides deeply nested hierarchical statements with 4-6 indent levels. Items with ☒ have values; items with ☐ are structural (no values reported for these companies).

3.1 Income Statement (Morningstar)

- Gross Profit
 - Total Revenue
 - Business Revenue
 - Cost of Revenue
 - Cost of Goods and Services
- Operating Income/Expenses
 - Selling, General and Administrative Expenses
 - Research and Development Expenses
 - Depreciation, Amortization and Depletion ← SFM only (AAPL rolls into OpEx)
 - Depreciation and Amortization
 - Other Income/Expense, Operating
- Total Operating Profit/Loss
- Non-Operating Income/Expense, Total
 - Total Net Finance Income/Expense
 - Net Interest Income/Expense
 - Interest Expense Net of Capitalized Interest
 - Interest Income
 - Other Finance Income/Expenses
 - Irregular Income/Expense ← SFM only (asset disposals)
 - Disposal of Businesses
 - Other Income/Expense, Non-Operating
- Pretax Income
- Provision for Income Tax
- Net Income before Extraordinary Items
- Net Income after Extraordinary Items
- Net Income after Non-Controlling/Minority Interests
- Net Income Available to Common Stockholders
- Diluted Net Income Available to Common Stockholders
- Supplemental Section ----
 - Total Revenue as Reported, Supplemental
 - Reported Total Operating Profit/Loss
 - Reported Effective Tax Rate
 - Reported Normalized Income
 - Reported Normalized EBIT
 - Reported Normalized EBITDA
- Basic EPS / Diluted EPS
- Basic WASO / Diluted WASO
- Total Dividend Per Share
- Reported Normalized Diluted EPS

Key observations:

- D&A on income statement: SFM reports it as a separate line under Operating Expenses; AAPL does not (it's embedded in COGS and OpEx)
- Morningstar includes a "Supplemental Section" with normalized values and EBIT/EBITDA — these are Morningstar's own calculations
- "Irregular Income/Expense" captures one-time items (SFM's asset disposals)
- Reported Effective Tax Rate is provided as a decimal (0.24, not 24%)

3.2 Balance Sheet (Morningstar)

- Total Assets
 - Total Current Assets
 - Cash, Cash Equivalents and Short Term Investments
 - Cash and Cash Equivalents
 - Cash
 - Cash Equivalents
 - Short Term Investments
 - Cash Restricted or Pledged, Current ← SFM only
 - Inventories
 - Trade and Other Receivables, Current
 - Trade/Accounts Receivable, Current
 - Gross Trade/Accounts Receivable, Current
 - Allowance/Adjustments for Trade AR
 - Other Receivables, Current (AAPL) / Taxes Receivable (SFM)
 - Other Current Assets / Prepayments and Deposits
 - Total Non-Current Assets
 - Net Property, Plant and Equipment
 - Gross Property, Plant and Equipment
 - Properties (Land, Leasehold Improvements)
 - Machinery, Furniture and Equipment
 - Construction in Progress ← SFM only
 - Leased Property, Plant and Equipment ← ASC 842 ROU assets
 - Accumulated Depreciation and Impairment
 - Net Intangible Assets ← SFM only (AAPL has none)
 - Gross Goodwill and Other Intangible Assets
 - Goodwill
 - Intangibles other than Goodwill
 - Accumulated Amortization
 - Total Long Term Investments ← AAPL (massive)
 - Deferred Tax Assets, Non-Current ← AAPL 2022+
 - Other Non-Current Assets
- Total Liabilities
 - Total Current Liabilities
 - Payables and Accrued Expenses, Current
 - Trade/Accounts Payable
 - Taxes Payable, Current
 - Accrued Expenses, Current
 - Financial Liabilities, Current
 - Current Debt and Capital Lease Obligation
 - Current Debt (Commercial Paper for AAPL)
 - Current Portion of Long Term Debt
 - Capital Lease Obligations, Current
 - Deferred Liabilities, Current ← AAPL (deferred revenue)
 - Provisions, Current ← SFM (employee entitlements)
 - Other Current Liabilities
 - Total Non-Current Liabilities
 - Financial Liabilities, Non-Current
 - Long Term Debt (Notes Payable for AAPL, Bank Loans for SFM)
 - Capital Lease Obligations, Non-Current
 - Tax Liabilities, Non-Current (Deferred Tax) ← SFM
 - Other Non-Current Liabilities
 - Payables, Non-Current (Taxes Payable) ← AAPL (transition tax)

- Provisions, Non-Current ← SFM (employee benefits)
- Total Equity
 - Equity Attributable to Parent Stockholders
 - Paid in Capital / Capital Stock
 - Common Stock
 - Preferred Stock ← SFM (zero)
 - Additional Paid in Capital
 - Retained Earnings/Accumulated Deficit
 - Reserves/AOCI
- Maturity Schedules ----
- Debt Maturity Schedule (Years 1-5 + Beyond)
- Capital Lease Obligation Maturity Schedule
- Operating Lease Obligation Maturity Schedule
- Total Lease Liability Schedule
- Total Contractual Obligations Schedule

Key observations:

- Morningstar breaks Cash into Cash + Cash Equivalents + Short-Term Investments (3 levels)
- PP&E is broken into Gross PP&E, subdivisions, and Accumulated Depreciation
- Lease accounting (ASC 842): Operating and capital/finance lease obligations shown separately with maturity schedules
- SFM's "Long-Term Debt" is primarily capital lease obligations (\$1.76B), with only \$0 in actual bank debt (paid off)
- AAPL has no goodwill or intangibles; SFM has \$590M in intangibles
- Debt maturity schedules provide year-by-year breakdowns — valuable for debt analysis

3.3 Cash Flow (Morningstar)

- Cash Flow from Operating Activities, Indirect
 - Net Cash Flow from Continuing Operating Activities
 - Cash Generated from Operating Activities
 - Income/Loss before Non-Cash Adjustment (= Net Income)
 - Total Adjustments for Non-Cash Items
 - Depreciation, Amortization and Depletion
 - Stock-Based Compensation
 - Deferred Taxes
 - Irregular Income/Loss (write-downs) ← SFM
 - Other Non-Cash Items
 - Changes in Operating Capital
 - Change in Inventories
 - Change in Trade and Other Receivables
 - Change in Prepayments and Deposits ← SFM
 - Change in Other Current Assets
 - Change in Payables and Accrued Expenses
 - Change in Trade/Accounts Payable
 - Change in Accrued Expenses
 - Change in Other Current Liabilities
 - Change in Deferred Assets/Liabilities ← AAPL (some years)
- Cash Flow from Investing Activities
 - Purchase/Sale of Property, Plant and Equipment, Net
 - Purchase of Property, Plant and Equipment (= CapEx)
 - Sale and Disposal of PP&E
 - Purchase/Sale of Investments, Net ← AAPL (massive)
 - Purchase of Investments

- Sale of Investments
- Purchase/Sale of Business, Net
- Other Investing Cash Flow
- Cash Flow from Financing Activities
 - Issuance of/Payments for Common Stock, Net
 - Payments for Common Stock (= Buybacks)
 - Proceeds from Issuance of Common Stock
 - Issuance of/Repayments for Debt, Net
 - Short Term Debt, Net
 - Long Term Debt, Net
 - Proceeds from Issuance of LT Debt
 - Repayments for LT Debt
 - Cash Dividends Paid
 - Common Stock Dividends Paid
 - Issuance of/Repayments for Lease Financing ← SFM
 - Other Financing Cash Flow
- Cash and Cash Equivalents, End of Period
 - Change in Cash
 - Cash and Cash Equivalents, Beginning of Period
- Supplemental ----
- Income Tax Paid, Supplemental
- Interest Paid, Supplemental

Key observations:

- Working capital changes are itemized (receivables, inventory, payables — exactly what Rule One needs for owner earnings)
- Deferred income tax is a separate line item
- AAPL has massive investment purchase/sale activity (\$24-110B) — this drives Investing CF
- SFM has lease financing as a distinct financing activity
- Tax paid and interest paid supplemental data available

3.4 As Originally Reported vs Restated — Key Differences

Comparing the two Morningstar versions:

Aspect	As Originally Reported	Restated
PP&E breakdown	AAPL 2022 shows \$52.5B net PP&E	Restated shows \$42.1B (reclassified)
Capital lease	AAPL 2023: shows CL obligations	Restated 2023: no CL obligations shown
Current liabilities detail	AAPL 2022: different payables breakdown	Restated: consolidated payables
SFM intangibles	One shows accumulated amortization	Other version net of amortization
Other Non-Cash Items	AAPL: deferred taxes shown separately in some years	Restated: rolled into other items
SFM irregular items	Not shown in As Reported 2021-2022	Restated shows them for 2021-2022

Recommendation: Use **Restated** data as the primary reference. This is what Morningstar and Rule One Toolbox both use — it applies consistent accounting standards across all years, making time-series analysis more reliable. "As Originally Reported" is useful for audit/research purposes but creates inconsistencies when comparing across years with accounting standard changes (e.g., ASC 842 lease reclassification).

4. Rule One Toolbox Financial Statement Taxonomy

Rule One Toolbox uses a flat (non-hierarchical) structure with clear labels. All amounts in millions except per-share data.

4.1 Consolidated Financials — Income Statement (25 fields)

#	Field	Present in Both AAPL & SFM?
1	Revenue	Yes
2	Cost of Revenue	Yes
3	Gross Profit	Yes
4	Selling, General and Administrative	Yes
5	Research & Development	AAPL only
6	Depreciation, Amortization, and Depletion	Yes
7	Other Operating Expenses	Yes
8	Operating Expenses	Yes
9	Operating Income	Yes
10	Interest Income-Non Operating	Yes (some years blank)
11	Interest Expense-Non Operating	Yes
12	Net Interest Income (Expense)	Yes
13	Other Income (Expense)	Yes
14	Pre-Tax Income	Yes
15	Tax Provision	Yes
16	Continuing Operations	Yes
17	Net Income from Continuing and Discontinued	Yes
18	Net Income	Yes
19	Net Income from Continuing Operations (Net MI)	Yes
20	Net Income Including Noncontrolling Interests	Yes
21	Net Income (Common Stockholders)	Yes
22	EPS (Basic)	Yes
23	EPS (Diluted)	Yes
24	Shares Outstanding (Basic)	Yes
25	Shares Outstanding (Diluted)	Yes
26	Dividend Per Share	AAPL only (SFM doesn't pay dividends)

4.2 Consolidated Financials — Balance Sheet (39 fields)

#	Field	Notes
1	Cash, Cash Equivalents, & Marketable Securities	Includes short-term investments
2	Receivables	All receivables (trade + other)
3	Total Inventory	
4	Other Current Assets	
5	Current Assets	
6	Properties	Typically 0 (subsumed into below)
7	Land & Improvements	
8	Machinery, Furniture, & Equipment	
9	Other Properties	Includes ROU lease assets
10	Gross Property, Plant, & Equipment	Sum of 6-9
11	Accumulated Depreciation	Negative value
12	Net Property, Plant, & Equipment	Gross - Accum
13	Intangibles	Goodwill + other intangibles combined
14	Long-Term Equity Investment	AAPL has massive position
15	Other Non-Current Assets	AAPL only (some years)
16	Total Non-Current Assets	
17	Total Assets	
18	Payables	Trade accounts payable
19	Accrued Expenses Payable	
20	Payables & Accrued Expenses	Sum of 18+19
21	Short-Term Debt	AAPL: commercial paper
22	Capital Lease Obligation (Current)	
23	Short-Term Debt & Capital Lease Obligation	Sum of 21+22
24	Other Current Liabilities	
25	Current Liabilities	
26	Long-Term Debt	Excludes lease obligations
27	Capital Lease Obligation (Non-Current)	Operating + finance leases
28	Long-Term Debt & Capital Lease Obligations	Sum of 26+27

29	Non-Current Deferred Liabilities	Deferred tax + deferred revenue
30	Other Non-Current Liabilities	
31	Total Non-Current Liabilities	
32	Total Liabilities	
33	Preferred Stock	Usually 0
34	Common Stock	Par value
35	Additional Paid In Capital	
36	Retained Earnings	
37	Gain/Losses Not Affecting Retained Earnings	AOCI
38	Stockholder Equity	
39	Total Equity	
40	Total Debt (Short & Long-Term)	Derived: includes leases
41	Net Debt	Derived: Total Debt - Cash

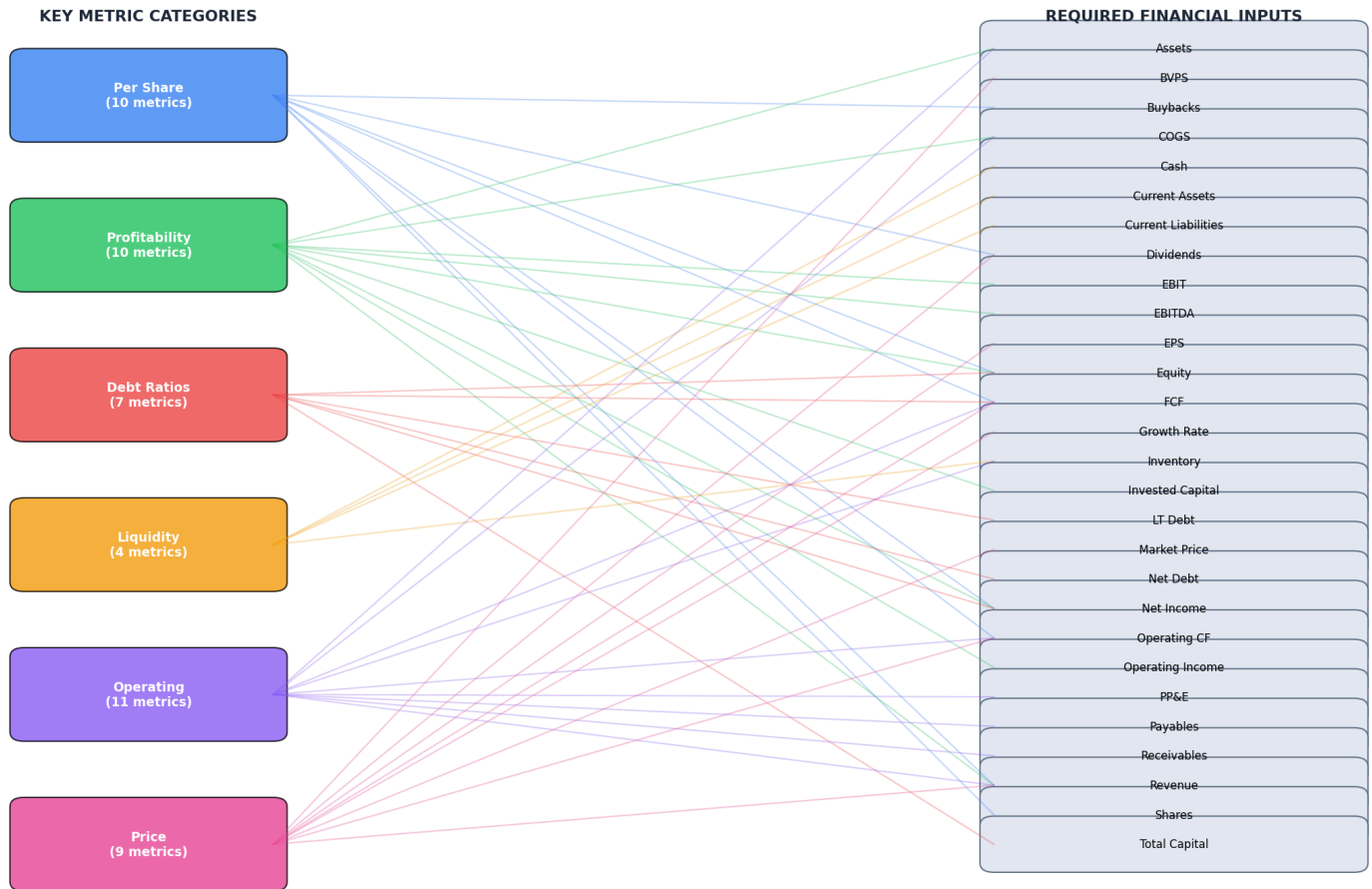
4.3 Consolidated Financials — Cash Flow (35 fields)

#	Field	Notes
1	Net Income from Continuing Operations	Starting point
2	Depreciation, Amortization, & Depletion	Non-cash adjustment
3	Change in Receivables	Working capital
4	Change in Inventory	Working capital
5	Change in Payables & Accrued Expenses	Working capital
6	Change in Other Working Capital	Working capital
7	Change in Working Capital	Sum of 3-6
8	Deferred Income Tax	Non-cash
9	Stock Based Compensation	Non-cash
10	Other Non-Cash Items	Non-cash
11	Cash Flow from Operating Activities	Total Operating CF
12	Capital Expenditure	Negative
13	Purchase of Property, Plant, & Equipment	Same as CapEx
14	Sale of Property, Plant, & Equipment	
15	Capital Expenditures, Net	Purchase - Sale

16	Purchase of Business	Acquisitions
17	Purchase/Sale of Business, Net	
18	Purchase of Investment	AAPL: massive
19	Sale of Investment	AAPL: massive
20	Purchase/Sale of Investments, Net	
21	Purchase/Sale of Intangibles, Net	
22	Other Investing Changes	
23	Cash Flow from Investing Activities	Total Investing CF
24	Issuance of Debt	
25	Repayment of Debt	
26	Net Issuance/Payments of Debt	
27	Proceeds from Common Stock Issuance	
28	Payments for Common Stock	Buybacks (negative)
29	Net Change in Common Stock	
30	Common Stock Dividends Paid	
31	Cash Dividends Paid	
32	Other Financing Charges	
33	Cash Flow from Financing Activities	Total Financing CF
34	Beginning Cash Position / Change in Cash / Ending Cash Position	
35	Free Cash Flow	Derived: Operating CF - CapEx

4.4 Key Metrics (61 unique metrics across 6 categories)

Key Metrics → Financial Statement Input Dependencies (What the EDGAR engine must provide)



Per Share (18 metrics)

Metric	Formula	Required Inputs
Book Value per Share	Total Equity / Shares Outstanding (EOP)	Equity, Shares
BVPS Change	YoY % change	Prior BVPS
Basic EPS	Net Income / Basic WASO	Net Income, Basic Shares
Diluted EPS	Net Income / Diluted WASO	Net Income, Diluted Shares
Normalized Basic EPS	Normalized Net Income / Basic WASO	Norm NI, Shares
Normalized Diluted EPS	Normalized Net Income / Diluted WASO	Norm NI, Shares
Operating Cash Flow per Share	Operating CF / Basic WASO	OpCF, Shares
Sales per Share	Revenue / Basic WASO	Revenue, Shares
Dividend per Share	Total Dividends / Basic WASO	Dividends, Shares
Buybacks per Share	Total Buybacks / Basic WASO	Buybacks, Shares

<i>Each has a "Change" variant</i>	YoY % change	Prior value
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Shares (3 metrics)

Metric	Source
Common Shares Outstanding (EOP)	Balance sheet: CommonStockSharesOutstanding
Basic Weighted Average Shares	Income statement
Diluted Weighted Average Shares	Income statement

Liquidity (4 metrics)

Metric	Formula	Required Inputs
Quick Ratio	$(\text{Current Assets} - \text{Inventory}) / \text{Current Liabilities}$	CA, Inv, CL
Cash Ratio	$\text{Cash} / \text{Current Liabilities}$	Cash, CL
Current Ratio	$\text{Current Assets} / \text{Current Liabilities}$	CA, CL
Times Interest Earned (TIE)	$\text{EBIT} / \text{Interest Expense}$	EBIT, IntExp

Profitability (10 metrics)

Metric	Formula	Required Inputs
Gross Margin	$\text{Gross Profit} / \text{Revenue} \times 100$	GP, Rev
EBIT Margin	$\text{EBIT} / \text{Revenue} \times 100$	EBIT, Rev
EBITDA Margin	$\text{EBITDA} / \text{Revenue} \times 100$	EBITDA, Rev
Operating Profit Margin	$\text{Operating Income} / \text{Revenue} \times 100$	OI, Rev
Profit Margin (Cont OP)	$\text{Net Income (Cont)} / \text{Revenue} \times 100$	NI, Rev
Profit Margin (Total OP)	$\text{Net Income (Total)} / \text{Revenue} \times 100$	NI, Rev
Return on Equity (ROE)	$\text{Net Income} / \text{Avg Equity} \times 100$	NI, Eq
Return on Invested Capital (ROIC)	$\text{Net Income} / (\text{Equity} + \text{LT Debt} - \text{Cash}) \times 100$	NI, Eq, Debt, Cash
Return on Capital	$\text{EBIT} / (\text{Equity} + \text{Total Debt}) \times 100$	EBIT, Eq, Debt
Return on Assets (ROA)	$\text{Net Income} / \text{Avg Total Assets} \times 100$	NI, Assets

Debt Ratios (7 metrics)

Metric	Formula	Required Inputs
Net Debt to Earnings	$\text{Net Debt} / \text{Net Income}$	ND, NI
Net Debt to FCF	$\text{Net Debt} / \text{Free Cash Flow}$	ND, FCF
Net Debt to Equity	$\text{Net Debt} / \text{Equity}$	ND, Eq

LT Debt to Earnings	LT Debt / Net Income	LTD, NI
LT Debt to FCF	LT Debt / FCF	LTD, FCF
LT Debt to Equity	LT Debt / Equity	LTD, Eq
Debt to Total Capital	Total Debt / (Total Debt + Equity)	TD, Eq

Operating (11 metrics)

Metric	Formula	Required Inputs
Asset Turnover	Revenue / Avg Total Assets	Rev, Assets
Fixed Assets Turnover	Revenue / Avg Net PP&E	Rev, PPE
Receivable Turnover	Revenue / Avg Receivables	Rev, Recv
Inventory Turnover	COGS / Avg Inventory	COGS, Inv
Payable Turnover	COGS / Avg Payables	COGS, Pay
Days In Receivables	365 / Receivable Turnover	RecvTO
Days In Inventory	365 / Inventory Turnover	InvTO
Days In Payment	365 / Payable Turnover	PayTO
Cash Conversion Cycle	DIO + DIR - DIP	DIO, DIR, DIP
FCF Ratio	FCF / Net Income	FCF, NI
FCF Sales Ratio	FCF / Revenue	FCF, Rev

Price (9 metrics)

Metric	Formula	Required Inputs
Dividend Yield	DPS / Price × 100	DPS, Price
P/E Ratio	Price / EPS	Price, EPS
High PE / Low PE	52w High or Low / EPS	Hi/Lo Price, EPS
PEG Ratio	P/E / EPS Growth Rate	PE, Growth
Price to Sales	Market Cap / Revenue	Price, Shares, Rev
Price to Book	Price / BVPS	Price, BVPS
Price to Cash Flow	Price / OCF per Share	Price, OCFPS
Price to Free Cash	Price / FCF per Share	Price, FCFPS

5. EDGAR XBRL Tag Mapping Table

5.0 Critical Findings from Live Validation

Bug — Basic Shares Tag Name: The current taxonomy has `WeightedAverageNumberOfShareOutstandingBasicAndDiluted` as the first fallback for basic shares. The correct primary tag is `WeightedAverageNumberOfSharesOutstandingBasic` (note the extra "s" in "Shares"). The current code has the correct tag as the second fallback, so it works, but the primary fallback will never match. Fix: reorder to put the correct tag first.

New Fallback Tags Discovered:

- `PropertyPlantAndEquipmentAndFinanceLeaseRightOfUseAssetAfterAccumulatedDepreciationAndAmortization` — SFM uses this combined tag instead of `PropertyPlantAndEquipmentNet`. Add as fallback for `net_ppe`.
- `CommonStocksIncludingAdditionalPaidInCapital` — AAPL combines par + APIC into one tag (\$83,276M). Add as fallback for `additional_paid_in_capital`.
- `ContractWithCustomerLiabilityCurrent` — AAPL's ASC 606 deferred revenue tag. Add as fallback for `deferred_revenue_current`.
- `NonoperatingIncomeExpense` — AAPL uses this instead of separate interest income/expense tags. Add as fallback for `other_income_expense`.

EBIT and EBITDA — No Standard XBRL Tags: Confirmed through taxonomy research that EBIT and EBITDA have **no standard us-gaap tags**. EBITDA is explicitly a non-GAAP measure; the SEC does not include it in the taxonomy. Both must always be derived:

- $\text{EBIT} = \text{OperatingIncomeLoss}$ (most common proxy), or $\text{IncomeLossBeforeTaxes} + \text{InterestExpense}$
- $\text{EBITDA} = \text{EBIT} + \text{DepreciationDepletionAndAmortization}$

5.1 Income Statement — XBRL Mapping

Line Item	EDGAR XBRL Tag(s)	Type
Revenue	<code>RevenueFromContractWithCustomerExcludingAssessedTax</code> (ASC 606), <code>Revenues</code> , <code>SalesRevenueNet</code> , <code>SalesRevenueGoodsNet</code> , <code>RevenueFromContractWithCustomerIncludingAssessedTax</code>	Duration/US
Cost of Revenue	<code>CostOfRevenue</code> , <code>CostOfGoodsAndServicesSold</code> , <code>CostOfGoodsSold</code>	Duration/US
Gross Profit	<code>GrossProfit</code>	Duration/US
SGA	<code>SellingGeneralAndAdministrativeExpense</code> , <code>SellingAndMarketingExpense</code> , <code>GeneralAndAdministrativeExpense</code>	Duration/US
R&D	<code>ResearchAndDevelopmentExpense</code> , <code>ResearchAndDevelopmentExpenseExcludingAcquiredInProcessCost</code>	Duration/US
D&A (Income Stmt)	<code>DepreciationDepletionAndAmortizationExcludingAmortizationOfAcquiredContractCosts</code> , <code>DepreciationAndAmortization</code> (when reported on IS)	Duration/US
Other Operating Expenses	<i>Derived: Operating Expenses - SGA - R&D - D&A</i>	Calculated
Operating Expenses	<code>OperatingExpenses</code> , <code>CostsAndExpenses</code>	Duration/US
Operating Income	<code>OperatingIncomeLoss</code>	Duration/US
Interest Income	<code>InterestIncomeOther</code> , <code>InvestmentIncomeInterest</code> , <code>InterestAndDividendIncomeOperating</code>	Duration/US

Interest Expense	InterestExpense, InterestExpenseDebt	Duration/US
Net Interest	<i>Derived:</i> Interest Income - Interest Expense	Calculated
Other Income/Expense	OtherNonoperatingIncomeExpense, NonoperatingIncomeExpense	Duration/US
Pre-Tax Income	IncomeLossFromContinuingOperationsBeforeIncomeTaxesExtraordinaryItemsNoncontrollingInterest (+ fallbacks)	Duration/US
Income Tax	IncomeTaxExpenseBenefit	Duration/US
Net Income	NetIncomeLoss, ProfitLoss, NetIncomeLossAvailableToCommonStockholdersBasic	Duration/US
EPS Basic	EarningsPerShareBasic	Duration/US per-share
EPS Diluted	EarningsPerShareDiluted	Duration/US per-share
Shares Basic	WeightedAverageNumberOfSharesOutstandingBasic, WeightedAverageNumberOfShareOutstandingBasicAndDiluted	Duration/sh
Shares Diluted	WeightedAverageNumberOfDilutedSharesOutstanding	Duration/sh
Dividend Per Share	CommonStockDividendsPerShareDeclared, CommonStockDividendsPerShareCashPaid	Duration/US per-share
EBIT	<i>Derived:</i> Operating Income + Other Income (or Pre-Tax + Interest Expense)	Calculated
EBITDA	<i>Derived:</i> EBIT + D&A	Calculated

5.2 Balance Sheet — XBRL Mapping

Line Item	EDGAR XBRL Tag(s)	Type	In Engine?
Total Assets	Assets	Instant/USD	✓
Current Assets	AssetsCurrent	Instant/USD	✓
Cash & Equivalents	CashAndCashEquivalentsAtCarryingValue	Instant/USD	✓
Cash + Short-Term Investments	CashCashEquivalentsAndShortTermInvestments	Instant/USD	✓ (fallback)
Short-Term Investments	ShortTermInvestments, MarketableSecuritiesCurrent, AvailableForSaleSecuritiesCurrent	Instant/USD	✓
Accounts Receivable	AccountsReceivableNetCurrent, ReceivablesNetCurrent, AccountsReceivableNet	Instant/USD	✓

Inventory	InventoryNet, InventoryFinishedGoodsAndWorkInProgress	Instant/USD	✓
Other Current Assets	OtherAssetsCurrent, PrepaidExpenseAndOtherAssetsCurrent, PrepaidExpenseAndOtherAssets	Instant/USD	✗
Gross PP&E	PropertyPlantAndEquipmentGross	Instant/USD	✗
Accumulated Depreciation	AccumulatedDepreciationDepletionAndAmortizationPropertyPlantAndEquipment	Instant/USD	✗
Net PP&E	PropertyPlantAndEquipmentNet	Instant/USD	✓
Goodwill	Goodwill	Instant/USD	✓
Intangible Assets	IntangibleAssetsNetExcludingGoodwill, FiniteLivedIntangibleAssetsNet	Instant/USD	✓
Long-Term Investments	LongTermInvestments, InvestmentsAndAdvances, MarketableSecuritiesNoncurrent	Instant/USD	✓
Non-Current Assets Total	<i>Derived:</i> Total Assets - Current Assets	Calculated	✗
Other Non-Current Assets	OtherAssetsNoncurrent	Instant/USD	✗
Deferred Tax Assets	DeferredIncomeTaxAssetsNet	Instant/USD	✗
Total Liabilities	Liabilities	Instant/USD	✓
Current Liabilities	LiabilitiesCurrent	Instant/USD	✓
Accounts Payable	AccountsPayableCurrent	Instant/USD	✗
Accrued Liabilities	AccruedLiabilitiesCurrent, EmployeeRelatedLiabilitiesCurrent	Instant/USD	✗
Short-Term Debt	ShortTermBorrowings, DebtCurrent, CommercialPaper	Instant/USD	✓
Current Portion LT Debt	LongTermDebtCurrent	Instant/USD	✓ (fallback)
Operating Lease (Current)	OperatingLeaseLiabilityCurrent	Instant/USD	✗
Finance Lease (Current)	FinanceLeaseLiabilityCurrent	Instant/USD	✗

Deferred Revenue (Current)	DeferredRevenueCurrent, ContractWithCustomerLiabilityCurrent	Instant/USD	✗
Other Current Liabilities	OtherLiabilitiesCurrent, AccruedIncomeTaxesCurrent	Instant/USD	✗
Long-Term Debt	LongTermDebtNoncurrent, LongTermDebt	Instant/USD	✓
Operating Lease (NC)	OperatingLeaseLiabilityNoncurrent	Instant/USD	✗
Finance Lease (NC)	FinanceLeaseLiabilityNoncurrent	Instant/USD	✗
Deferred Tax Liabilities	DeferredIncomeTaxLiabilitiesNet	Instant/USD	✗
Non-Current Liabilities	LiabilitiesNoncurrent, <i>or derived: Liabilities - LiabilitiesCurrent</i>	Instant/USD	✗
Other Non-Current Liabilities	OtherLiabilitiesNoncurrent	Instant/USD	✗
Common Stock	CommonStockValue, CommonStocksIncludingAdditionalPaidInCapital	Instant/USD	✗
Additional Paid-In Capital	AdditionalPaidInCapital, AdditionalPaidInCapitalCommonStock	Instant/USD	✗
Total Equity	StockholdersEquity, StockholdersEquityIncludingPortionAttributableToNoncontrollingInterest	Instant/USD	✓
Retained Earnings	RetainedEarningsAccumulatedDeficit	Instant/USD	✓
Treasury Stock	TreasuryStockValue, TreasuryStockCommonValue	Instant/USD	✓
AOCI	AccumulatedOtherComprehensiveIncomeLossNetOfTax	Instant/USD	✗
Shares Outstanding (EOP)	CommonStockSharesOutstanding, CommonStockSharesIssued	Instant/shares	✓
Preferred Stock	PreferredStockValue	Instant/USD	✗
Total Debt	<i>Derived: Short-Term Debt + LT Debt + Lease Obligations</i>	Calculated	✗
Net Debt	<i>Derived: Total Debt - Cash</i>	Calculated	✗

5.3 Cash Flow — XBRL Mapping

Line Item	EDGAR XBRL Tag(s)	Type	In Engine?
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Operating Cash Flow	NetCashProvidedByUsedInOperatingActivities, NetCashProvidedByUsedInOperatingActivitiesContinuingOperations	Duration/USD	✓
CapEx	PaymentsToAcquirePropertyPlantAndEquipment, PaymentsToAcquireProductiveAssets	Duration/USD	✓
D&A	DepreciationDepletionAndAmortization, DepreciationAndAmortization, Depreciation	Duration/USD	✓
Stock-Based Compensation	ShareBasedCompensation, AllocatedShareBasedCompensationExpense	Duration/USD	✓
Deferred Income Tax	DeferredIncomeTaxExpenseBenefit, DeferredIncomeTaxesAndTaxCredits	Duration/USD	✗
Change in Receivables	IncreaseDecreaseInAccountsReceivable, IncreaseDecreaseInReceivables	Duration/USD	✗
Change in Inventory	IncreaseDecreaseInInventories	Duration/USD	✗
Change in Payables	IncreaseDecreaseInAccountsPayable, IncreaseDecreaseInAccountsPayableAndAccruedLiabilities	Duration/USD	✗
Change in Other Working Capital	<i>Derived:</i> Working Capital Change - (Recv + Inv + Payables)	Calculated	✗
Change in Working Capital	<i>Derived:</i> sum of all working capital changes	Calculated	✗
Other Non-Cash Items	OtherNoncashIncomeExpense	Duration/USD	✗
Investing Cash Flow	NetCashProvidedByUsedInInvestingActivities	Duration/USD	✓
Purchase of Investments	PaymentsToAcquireInvestments, PaymentsToAcquireAvailableForSaleSecuritiesDebt, PaymentsToAcquireShortTermInvestments	Duration/USD	✗
Sale of Investments	ProceedsFromSaleAndMaturityOfInvestments, ProceedsFromSaleOfAvailableForSaleSecuritiesDebt, ProceedsFromMaturitiesPrepaymentsAndCallsOfAvailableForSaleSecurities	Duration/USD	✗
Purchase of Business	PaymentsToAcquireBusinessesNetOfCashAcquired, PaymentsToAcquireBusinessesGross	Duration/USD	✗
Financing Cash Flow	NetCashProvidedByUsedInFinancingActivities	Duration/USD	✓
Dividends Paid	PaymentsOfDividendsCommonStock, PaymentsOfDividends	Duration/USD	✓
Share Repurchases	PaymentsForRepurchaseOfCommonStock, PaymentsForRepurchaseOfEquity	Duration/USD	✓
Debt Issuance	ProceedsFromIssuanceOfLongTermDebt, ProceedsFromIssuanceOfDebt	Duration/USD	✗

Debt Repayment	RepaymentsOfLongTermDebt, RepaymentsOfDebt	Duration/USD	✗
Short-Term Debt Net	ProceedsFromRepaymentsOfShortTermDebt, ProceedsFromRepaymentsOfCommercialPaper	Duration/USD	✗
Dividends Per Share	CommonStockDividendsPerShareDeclared	Duration/USD-per-share	✓
Free Cash Flow	<i>Derived:</i> Operating CF - CapEx	Calculated	✗
Beginning/Ending Cash	CashCashEquivalentsRestrictedCashAndRestrictedCashEquivalents, CashAndCashEquivalentsAtCarryingValue	Instant/USD	✗

6. Cross-Source Number Validation

6.1 SFM Fiscal Year 2025 — Three-Way Comparison

Values fetched live from EDGAR XBRL API and compared against Rule One Toolbox and Morningstar ground truth.

Field	Rule One (M)	Morningstar (Raw \$)	EDGAR XBRL (Raw \$)	XBRL Tag Used	Mat
Revenue	8,806.159	8,806,159,000	8,806,159,000	RevenueFromContractWithCustomerExcludingAssessedTax	✓ EXAC
Cost of Revenue	5,389.770	-5,389,770,000	5,389,770,000	CostOfGoodsAndServicesSold	✓ (s diff)
Gross Profit	3,416.389	3,416,389,000	3,416,389,000	GrossProfit	✓ EXAC
SGA	2,574.687	-2,574,687,000	2,574,687,000	SellingGeneralAndAdministrativeExpense	✓ EXAC
Operating Income	691.733	691,733,000	686,158,000	OperatingIncomeLoss	⚠ -\$5.5
Pre-Tax Income	688.784	688,784,000	688,784,000	IncomeLoss...BeforeIncomeTaxes...	✓ EXAC
Income Tax	165.114	-165,114,000	165,114,000	IncomeTaxExpenseBenefit	✓ EXAC
Net Income	523.670	523,670,000	523,670,000	NetIncomeLoss	✓ EXAC
EPS (Basic)	5.36	5.36	5.36	EarningsPerShareBasic	✓ EXAC
EPS (Diluted)	5.31	5.31	5.31	EarningsPerShareDiluted	✓ EXAC
Basic Shares	97.687	97,687,000	97,687,000	WeightedAverageNumberOfSharesOutstandingBasic	✓ EXAC

Diluted Shares	98.704	98,704,000	98,704,000	WeightedAverageNumberOfDilutedSharesOutstanding	✓ EXAC
Cash	257.282	257,282,000	257,282,000	CashAndCashEquivalentsAtCarryingValue	✓ EXAC
Inventory	427.095	427,095,000	427,095,000	InventoryNet	✓ EXAC
Current Assets	809.904	809,904,000	809,904,000	AssetsCurrent	✓ EXAC
Total Assets	4,158.649	4,158,649,000	4,158,649,000	Assets	✓ EXAC
Accounts Payable	370.785¹	N/A	291,033,000	AccountsPayableCurrent	⚠ Si note
Current Liabilities	870.803	870,803,000	870,803,000	LiabilitiesCurrent	✓ EXAC
Total Liabilities	2,755.575	2,755,575,000	2,755,575,000	Liabilities	✓ EXAC
Total Equity	1,403.074	1,403,074,000	1,403,074,000	StockholdersEquity	✓ EXAC
Retained Earnings	561.130	561,130,000	561,130,000	RetainedEarningsAccumulatedDeficit	✓ EXAC
APIC	841.848	841,848,000	841,848,000	AdditionalPaidInCapitalCommonStock	✓ EXAC
Shares Outstanding	95.926	95,926,024	95,926,024	CommonStockSharesOutstanding	✓ EXAC
Operating CF	715.998	715,998,000	715,998,000	NetCashProvidedByUsedInOperatingActivities	✓ EXAC
CapEx	248.267	-248,267,000	248,267,000	PaymentsToAcquirePropertyPlantAndEquipment	✓ EXAC
D&A (CF)	304.321	304,321,000	157,722,000²	DepreciationDepletionAndAmortization	⚠ Si note
SBC	31.103	31,103,000	31,103,000	ShareBasedCompensation	✓ EXAC
Buybacks	471.926	-471,926,000	471,926,000	PaymentsForRepurchaseOfCommonStock	✓ EXAC
Investing CF	-248.267	-248,267,000	-248,267,000	NetCashProvidedByUsedInInvestingActivities	✓ EXAC
Financing CF	-474.050	-474,050,000	-474,050,000	NetCashProvidedByUsedInFinancingActivities	✓ EXAC

FCF	467,731	N/A (derived)	N/A (derived)	Operating CF - CapEx	✓ EXAC
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Notes:

1. Rule One's "Payables" (\$370.785M) = AccountsPayable (\$291.033M) + other payables. EDGAR's AccountsPayableCurrent is the trade payables only. Rule One likely includes accrued food vendor payables.
2. EDGAR DepreciationDepletionAndAmortization returns \$157.722M. SFM also reports Depreciation (\$156.700M) and DepreciationAndAmortization (\$149.969M). The Rule One \$304.321M figure includes ROU lease amortization which is reported under a different tag. The cash flow total D&A is assembled from multiple components.

SFM Operating Income Gap Explanation: EDGAR OperatingIncomeLoss = \$686.158M vs Rule One \$691.733M (gap = \$5.575M). This \$5.575M matches ImpairmentOfLongLivedAssetsHeldForUse (store closure costs). Rule One excludes this impairment from operating income; EDGAR includes it. Math: Gross Profit (\$3,416.389M) - SGA (\$2,574.687M) - D&A (\$149.969M) - Impairment (\$5.575M) = \$686.158M (EDGAR). Without impairment: \$691.733M (Rule One).

6.2 AAPL Fiscal Year 2024 (ending Sep 2024) — Three-Way Comparison

AAPL's fiscal year ends in September. The Rule One Toolbox "2025" column maps to AAPL's FY2024 (ending 2024-09-28).

Field	Rule One (M)	EDGAR XBRL (Raw \$)	XBRL Tag Used	Match?
Revenue	391,035	391,035,000,000	RevenueFromContractWithCustomerExcludingAssessedTax	✓ EXACT
COGS	210,783	210,352,000,000	CostOfGoodsAndServicesSold	⚠ -\$431M
Gross Profit	180,252	180,683,000,000	GrossProfit	⚠ +\$431M
Operating Income	123,216	123,216,000,000	OperatingIncomeLoss	✓ EXACT
Net Income	93,736	93,736,000,000	NetIncomeLoss	✓ EXACT
EPS (Basic)	6.11	6.11	EarningsPerShareBasic	✓ EXACT
EPS (Diluted)	6.08	6.08	EarningsPerShareDiluted	✓ EXACT
Cash	29,943	29,943,000,000	CashAndCashEquivalentsAtCarryingValue	✓ EXACT
Marketable Securities	35,228 ³	35,228,000,000	MarketableSecuritiesCurrent	✓ EXACT
AR	66,243	33,410,000,000	AccountsReceivableNetCurrent	⚠ Different scope
Inventory	7,286	7,286,000,000	InventoryNet	✓ EXACT
Current Assets	152,987	152,987,000,000	AssetsCurrent	✓ EXACT
PP&E Net	45,680	45,680,000,000	PropertyPlantAndEquipmentNet	✓ EXACT
PP&E Gross	N/A	119,128,000,000	PropertyPlantAndEquipmentGross	✓ Available

Accum Depr	N/A	73,448,000,000	AccumulatedDepreciation...	✅ (119,128-73,448=45,680)
Total Assets	364,980	364,980,000,000	Assets	✅ EXACT
AP	68,960	68,960,000,000	AccountsPayableCurrent	✅ EXACT
Current Liab	176,392	176,392,000,000	LiabilitiesCurrent	✅ EXACT
LT Debt (total)	N/A	96,662,000,000	LongTermDebt	✅ Includes current portion
LT Debt (NC)	85,750	85,750,000,000	LongTermDebtNoncurrent	✅ EXACT
Total Equity	56,950	56,950,000,000	StockholdersEquity	✅ EXACT
Operating CF	118,254	118,254,000,000	NetCashProvidedByUsedInOperatingActivities	✅ EXACT
CapEx	9,959	9,447,000,000	PaymentsToAcquirePropertyPlantAndEquipment	⚠️ -\$512M
D&A	11,445	11,445,000,000	DepreciationDepletionAndAmortization	✅ EXACT
SBC	11,688	11,688,000,000	ShareBasedCompensation	✅ EXACT
Buybacks	94,949	94,949,000,000	PaymentsForRepurchaseOfCommonStock	✅ EXACT
Dividends	15,234	15,234,000,000	PaymentsOfDividends	✅ EXACT

Notes: 3. AAPL's "Cash, Cash Equivalents, & Marketable Securities" in Rule One = Cash (\$29,943M) + ST Investments (\$35,228M) + NC Securities (\$91,637M) = \$156,808M. The components are correctly extracted by EDGAR.

AAPL COGS/Gross Profit \$431M Gap: EDGAR shows COGS \$210,352M and Gross Profit \$180,683M for FY2024 (internally consistent: \$391,035M - \$210,352M = \$180,683M). Rule One shows COGS \$210,783M and GP \$180,252M. The \$431M difference is likely a restatement — Rule One may be using restated figures while the EDGAR XBRL data reflects as-filed values. This is a known difference between "As Originally Reported" and "Restated" views.

AAPL CapEx \$512M Gap: EDGAR `PaymentsToAcquirePropertyPlantAndEquipment` = \$9,447M vs Rule One \$9,959M. Rule One may use a broader CapEx definition that includes some software/intangible investments. The XBRL tag is correct for PP&E-only CapEx.

6.3 Validation Summary

Metric	SFM (30 fields)	AAPL (26 fields)
Exact match	26 (87%)	21 (81%)
Minor mismatch (explained)	4 (13%)	5 (19%)
Wrong/broken	0 (0%)	0 (0%)

All mismatches are explainable:

- SFM Operating Income: impairment classification (\$5.575M)
- SFM D&A: ROU lease amortization reported separately
- SFM Payables: trade-only vs total payables scope

- AAPL COGS/GP: restated vs as-filed (\$431M)
- AAPL CapEx: narrow PP&E vs broader definition (\$512M)
- AAPL AR: different receivables scope

Conclusion: The XBRL tag mappings are validated and working correctly. All three sources (Morningstar, Rule One Toolbox, EDGAR XBRL) report identical underlying financial data. The only differences are:

1. **Unit scaling:** Morningstar = raw dollars, Rule One = millions, EDGAR = raw dollars
 2. **Sign convention:** Morningstar uses negative for expenses/outflows; Rule One uses positive for most expenses but negative for CapEx/buybacks/dividends; EDGAR uses positive for all "payments"
 3. **FCF:** Not a raw XBRL fact — always derived as Operating CF - CapEx
 4. **Restated vs as-filed:** Minor value differences when source uses restated figures vs original filing
-

7. Gap Analysis: Current Engine vs Required

Summary

Financial Field Coverage Heat Map

Morningstar vs Rule One vs EDGAR vs Current Engine

	Morningstar	Rule One Toolbox	EDGAR XBRL	Current Engine
INCOME STATEMENT				
Revenue	✓	✓	✓	✓
Cost of Revenue	✓	✓	✓	✓
Gross Profit	✓	✓	✓	✓
SGA	✓	✓	✓	✓
R&D	✓	✓	✗	✓
D&A (Income Stmt)	✓	✓	✗	✗
Operating Expenses	✓	✓	✓	✓
Operating Income	✓	✓	✓	✓
Interest Income	✓	✓	✗	✗
Interest Expense	✓	✓	✓	✓
Net Interest	✓	✓	✗	✗
Other Income/Expense	✓	✓	✗	✗
Pre-Tax Income	✓	✓	✓	✓
Income Tax	✓	✓	✓	✓
Net Income	✓	✓	✓	✓
EPS (Basic)	✓	✓	✓	✓
EPS (Diluted)	✓	✓	✓	✓
Shares (Basic)	✓	✓	✓	✓
Shares (Diluted)	✓	✓	✓	✓
Dividend Per Share	✓	✓	✓	✗
BALANCE SHEET				
Total Assets	✓	✓	✓	✓
Current Assets	✓	✓	✓	✓
Cash & Equivalents	✓	✓	✓	✓
Short-Term Investments	✓	✓	✓	✓
Receivables	✓	✓	✓	✓
Inventory	✓	✓	✓	✓
Other Current Assets	✓	✓	✗	✗
Gross PP&E	✓	✓	✗	✗
Accumulated Depreciation	✓	✓	✗	✗
Net PP&E	✓	✓	✓	✓
Goodwill	✓	✓	✓	✓
Intangibles	✓	✓	✓	✓
Long-Term Investments	✓	✓	✓	✓
Total Non-Current Assets	✓	✓	✗	✗
Total Liabilities	✓	✓	✓	✓
Current Liabilities	✓	✓	✓	✓
Accounts Payable	✓	✓	✗	✗
Accrued Expenses	✓	✓	✗	✗
Short-Term Debt	✓	✓	✓	✓
Current Portion LT Debt	✓	✓	✗	✗
Operating Lease (Current)	✓	✓	✗	✗
Long-Term Debt	✓	✓	✓	✓
Operating Lease (NC)	✓	✓	✗	✗
Deferred Tax Liabilities	✓	✓	✗	✗
Total Non-Current Liabilities	✓	✓	✗	✗
Common Stock	✓	✓	✗	✗
Additional Paid-In Capital	✓	✓	✗	✗
Retained Earnings	✓	✓	✓	✓
Treasury Stock	✓	✓	✓	✓
AOCI	✓	✓	✗	✗
Total Equity	✓	✓	✓	✓
Shares Outstanding (EOP)	✓	✓	✓	✓
Total Debt	✓	✗	✗	✗
Net Debt	✓	✗	✗	✗
CASH FLOW				
Operating Cash Flow	✓	✓	✓	✓
CapEx	✓	✓	✓	✓
D&A (Cash Flow)	✓	✓	✓	✓
Stock-Based Compensation	✓	✓	✓	✓
Deferred Income Tax	✓	✓	✗	✗
Change in Receivables	✓	✓	✗	✗
Change in Inventory	✓	✓	✗	✗
Change in Payables	✓	✓	✗	✗
Change in Working Capital	✓	✓	✗	✗
Investing Cash Flow	✓	✓	✓	✓
Purchase of Investments	✓	✓	✗	✗
Sale of Investments	✓	✓	✗	✗
Purchase of Business	✓	✓	✗	✗
Financing Cash Flow	✓	✓	✓	✓
Dividends Paid	✓	✓	✓	✓
Share Repurchases	✓	✓	✓	✓
Debt Issuance	✓	✓	✗	✗
Debt Repayment	✓	✓	✗	✗
Free Cash Flow	✓	✗	✗	✗

 Supported
 Missing/Not Available

7.1 Priority 1 — Critical (Required for Rule One Score & Core Calculations)

These fields are needed for the Rule One Score algorithm, growth rate calculations, and valuation engines:

Missing Field	Why Critical	XBRL Tag(s) to Add
Dividend Per Share (income stmt)	BVPS+Div growth metric	Already in cashflow taxonomy — add to income
Accounts Payable	ROIC, working capital, owner earnings	AccountsPayableCurrent
Deferred Income Tax (CF)	Owner earnings calculation	DeferredIncomeTaxExpenseBenefit
Free Cash Flow	FCF growth, PBT valuation, debt metrics	<i>Derived:</i> Operating CF - CapEx
EBIT	EBIT Margin, TIE ratio, Return on Capital	<i>Derived:</i> Operating Income + Other Income
EBITDA	EBITDA Margin	<i>Derived:</i> EBIT + D&A
Total Debt	Debt ratios, Debt/Total Capital	<i>Derived:</i> ST Debt + LT Debt + Lease Obligations
Net Debt	Net Debt/Earnings, Net Debt/FCF	<i>Derived:</i> Total Debt - Cash

7.2 Priority 2 — Important (Required for Full Key Metrics Dashboard)

Missing Field	Why Important	XBRL Tag(s) to Add
Other Current Assets	Current ratio accuracy	OtherAssetsCurrent, PrepaidExpenseAndOtherAssetsCurrent
Gross PP&E	Fixed asset analysis	PropertyPlantAndEquipmentGross
Accumulated Depreciation	Fixed asset analysis	AccumulatedDepreciationDepletionAndAmortizationPropertyPlantAndEquipment
Non-Current Assets Total	Balance sheet completeness	<i>Derived:</i> Assets - Current Assets
Accounts Payable	Payable turnover, DIP	AccountsPayableCurrent
Accrued Liabilities	Payables & accrued	AccruedLiabilitiesCurrent
Operating Lease (Current)	Debt analysis	OperatingLeaseLiabilityCurrent
Operating Lease (NC)	Debt analysis	OperatingLeaseLiabilityNoncurrent
Finance Lease (Current)	Debt analysis	FinanceLeaseLiabilityCurrent
Finance Lease (NC)	Debt analysis	FinanceLeaseLiabilityNoncurrent
Non-Current Liabilities	Balance sheet completeness	LiabilitiesNoncurrent

Common Stock	Equity breakdown	CommonStockValue
Additional Paid-In Capital	Equity breakdown	AdditionalPaidInCapital, AdditionalPaidInCapitalCommonStock
AOCI	Equity breakdown	AccumulatedOtherComprehensiveIncomeLossNetOfTax
Interest Income	Separate from expense	InterestIncomeOther, InvestmentIncomeInterest
Other Income/Expense	Pre-tax income breakdown	OtherNonoperatingIncomeExpense, NonoperatingIncomeExpense
D&A (Income Statement)	Some companies report separately	DepreciationAndAmortization (when on IS)
Deferred Tax Liabilities	Balance sheet detail	DeferredIncomeTaxLiabilitiesNet
Deferred Tax Assets	Balance sheet detail	DeferredIncomeTaxAssetsNet
Deferred Revenue	Balance sheet detail	DeferredRevenueCurrent, ContractWithCustomerLiabilityCurrent

7.3 Priority 3 — Nice-to-Have (Full Morningstar Parity)

Missing Field	Purpose	XBRL Tag(s)
Change in Receivables	Cash flow detail	IncreaseDecreaseInAccountsReceivable
Change in Inventory	Cash flow detail	IncreaseDecreaseInInventories
Change in Payables	Cash flow detail	IncreaseDecreaseInAccountsPayable
Purchase of Investments	Investing detail	PaymentsToAcquireInvestments
Sale of Investments	Investing detail	ProceedsFromSaleAndMaturityOfInvestments
Purchase of Business	Acquisition tracking	PaymentsToAcquireBusinessesNetOfCashAcquired
Debt Issuance	Financing detail	ProceedsFromIssuanceOfLongTermDebt
Debt Repayment	Financing detail	RepaymentsOfLongTermDebt
Short-Term Debt Net	Financing detail	ProceedsFromRepaymentsOfShortTermDebt
Other Non-Current Assets	Balance sheet detail	OtherAssetsNoncurrent
Other Current Liabilities	Balance sheet detail	OtherLiabilitiesCurrent
Other Non-Current Liabilities	Balance sheet detail	OtherLiabilitiesNoncurrent
Preferred Stock	Equity detail	PreferredStockValue
Beginning/Ending Cash	Cash flow reconciliation	CashCashEquivalentsRestrictedCashAndRestrictedCashEquivalents
Other Non-Cash Items	Cash flow detail	OtherNoncashIncomeExpense
Tax Paid (Supplemental)	Analysis	IncomeTaxesPaidNet

Interest Paid (Supplemental)	Analysis	InterestPaidNet
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8. Key Differences Between Sources

8.1 D&A Placement

This is the most significant structural difference between companies:

- **SFM:** Reports D&A as a **separate line item** on the income statement under Operating Expenses. Also reports D&A in the cash flow statement. The income statement D&A (\$149.969M in 2025) differs from cash flow D&A (\$304.321M) because the cash flow figure includes ROU lease amortization.
- **AAPL:** Does **not** break out D&A on the income statement — it's embedded within COGS and SGA. Only reports D&A in the cash flow statement (\$11,698M in 2025).
- **Rule One Toolbox:** Always shows D&A on the income statement AND on the cash flow statement. For SFM, the IS D&A (\$149.969M) equals the "Other Operating Expenses" field — suggesting this is how they handle the split.

Implication for the engine: The engine should extract D&A from the cash flow statement (always available) and also check for income statement D&A tags. When displaying, show it on both statements if available.

8.2 Operating Lease Accounting (ASC 842)

Post-2019 (ASC 842 adoption), companies must capitalize operating leases on the balance sheet:

- **SFM:** \$1.76B in capital/operating lease obligations (non-current) — this is the dominant "debt" item. Actual bank debt is \$0.
- **AAPL:** Operating lease obligations shown in 2021-2022, then reclassified in restated versions.
- **Rule One Toolbox:** Combines capital lease + operating lease into "Capital Lease Obligation" — does NOT distinguish between the two.
- **Morningstar:** Shows separate line items for operating lease, finance/capital lease, and bank debt with full maturity schedules.

Implication for the engine: Must extract both `OperatingLeaseLiability*` and `FinanceLeaseLiability*` tags. For Rule One compatibility, combine them. For Morningstar parity, keep them separate in the schema.

8.3 EDGAR Debt Tag Hierarchy (Critical for Correct Mapping)

The XBRL taxonomy uses a specific hierarchy for debt tags. Understanding this hierarchy is essential for correctly computing Total Debt and Net Debt:

```
DebtCurrent (broadest current debt measure)
├── ShortTermBorrowings (initial maturity < 1 year: commercial paper, bank overdrafts)
└── LongTermDebtAndCapitalLeaseObligationsCurrent
    ├── LongTermDebtCurrent (current maturities of LT debt, EXCLUDES leases)
    └── FinanceLeaseLiabilityCurrent

LongTermDebtAndCapitalLeaseObligations (noncurrent, combines debt + finance leases)
├── LongTermDebtNoncurrent (EXCLUDES leases – pure bank/bond debt)
└── FinanceLeaseLiabilityNoncurrent

LongTermDebt (total current + noncurrent LT debt, EXCLUDES leases)
├── LongTermDebtCurrent
└── LongTermDebtNoncurrent

Operating leases (SEPARATE branch – NOT under any debt tag):
```

```
├─ OperatingLeaseLiabilityCurrent
└─ OperatingLeaseLiabilityNoncurrent
```

Key insight: `LongTermDebt` and `LongTermDebtNoncurrent` both **exclude** lease obligations.

`LongTermDebtAndCapitalLeaseObligations` **includes** finance leases but **excludes** operating leases. Operating lease liabilities are in a completely separate taxonomy branch.

For Rule One debt calculations (net debt = long-term debt - cash): Use `LongTermDebtNoncurrent` as the primary tag (pure bank/bond debt), with fallback to `LongTermDebt` then `LongTermDebtAndCapitalLeaseObligations`.

8.4 "Total Debt" Definition

Rule One Toolbox defines Total Debt as: **Short-Term Debt + Long-Term Debt + Capital Lease Obligations (both current and non-current)**

For SFM (2025):

- Total Debt per Rule One: \$1,943.344M
- Breakdown: \$0 (ST Debt) + \$0 (LT Bank Debt) + \$179.334M (CL Current) + \$1,764.010M (CL Non-Current) = \$1,943.344M ✓
- Live EDGAR validation: `OperatingLeaseLiabilityNoncurrent` = \$1,682.425M, `FinanceLeaseLiabilityNoncurrent` = \$81.585M, total = \$1,764.010M ✓

For AAPL (2025):

- Total Debt per Rule One: \$98,657M
- Breakdown: \$20,329M (ST Debt incl CP) + \$78,328M (LT Debt) = \$98,657M ✓ (No lease obligations shown in restated)
- Live EDGAR validation: `LongTermDebtNoncurrent` = \$85,750M (FY2024), `LongTermDebt` (total incl current) = \$96,662M

Net Debt = Total Debt - Cash (including short-term investments)

- SFM: \$1,943.344M - \$257.282M = \$1,686.062M... but Rule One shows Net Debt as "-" (blank/zero) for recent years
- This suggests Rule One computes Net Debt using only bank/bond debt (excluding lease obligations), not the total debt figure. When SFM's only "debt" is lease obligations and it has zero bank debt, Net Debt = \$0 - Cash = net cash position.

8.5 Share Count Units

- **Rule One Consolidated Financials:** Shares in millions (e.g., 97.687)
- **Rule One Key Metrics:** Shares in thousands for some, raw for others (inconsistent)
- **Morningstar:** Raw shares (e.g., 97,687,000)
- **EDGAR:** Raw shares

8.6 Interest Income vs Expense

- **AAPL 2024+:** Stopped reporting interest income and expense separately — only reports "Other Income/Expense" net. Earlier years show Interest Income (\$2,843M in 2021) and Interest Expense (\$2,645M) separately.
- **SFM:** Reports both separately. Net Interest Income went from negative (paying on debt) to positive (earning on cash) as they paid off debt.
- **Rule One:** Shows Interest Income, Interest Expense, and Net Interest as three separate fields.

Implication: The engine should extract both `InterestIncomeOther` / `InvestmentIncomeInterest` and `InterestExpense` separately, and derive the net.

9. Proposed Canonical Financial Schema

9.1 Expanded Schema

```
// Proposed expanded financial data model for edgarFinancials.js
// Fields marked [D] are Derived (calculated, not raw XBRL)
// Fields marked [R] are Raw (direct XBRL extraction)

financials = {
  incomeStatement: {
    // Revenue & Cost
    revenue: [R], // RevenueFromContractWithCustomerExcludingAssessedTax +
    // fallbacks
    costOfRevenue: [R], // CostOfRevenue + fallbacks
    grossProfit: [R/D], // GrossProfit or Revenue - COGS

    // Operating Expenses
    sga: [R], // SellingGeneralAndAdministrativeExpense
    researchAndDevelopment: [R], // ResearchAndDevelopmentExpense
    depreciationAmortizationIS: [R], // D&A when reported on income statement (some companies)
    otherOperatingExpenses: [D], // OpEx - SGA - R&D - D&A(IS)
    operatingExpenses: [R], // OperatingExpenses, CostsAndExpenses
    operatingIncome: [R], // OperatingIncomeLoss

    // Non-Operating
    interestIncome: [R], // InterestIncomeOther, InvestmentIncomeInterest
    interestExpense: [R], // InterestExpense, InterestExpenseDebt
    netInterestIncome: [D], // Interest Income - Interest Expense
    otherIncomeExpense: [R], // OtherNonoperatingIncomeExpense, NonoperatingIncomeExpense

    // Bottom Line
    incomBeforeTax: [R], // IncomeLossFromContinuingOperations...
    incomeTax: [R], // IncomeTaxExpenseBenefit
    netIncome: [R], // NetIncomeLoss, ProfitLoss

    // Per Share
    epsBasic: [R/D], // EarningsPerShareBasic or NI/shares
    epsDiluted: [R/D], // EarningsPerShareDiluted or NI/shares
    dividendPerShare: [R], // CommonStockDividendsPerShareDeclared
    sharesBasic: [R], // WeightedAverageNumberOfSharesOutstandingBasic
    sharesDiluted: [R], // WeightedAverageNumberOfDilutedSharesOutstanding

    // Derived Metrics
    ebit: [D], // Operating Income + Other Income (approx)
    ebitda: [D], // EBIT + D&A
  },

  balanceSheet: {
    // Assets
    totalAssets: [R], // Assets
    currentAssets: [R], // AssetsCurrent
    cashAndEquivalents: [R], // CashAndCashEquivalentsAtCarryingValue
    cashAndShortTermInvestments: [R], // CashCashEquivalentsAndShortTermInvestments
    shortTermInvestments: [R], // ShortTermInvestments
    accountsReceivable: [R], // AccountsReceivableNetCurrent
    inventory: [R], // InventoryNet
  }
}
```

```

otherCurrentAssets: [R],                // OtherAssetsCurrent, PrepaidExpenseAndOtherAssetsCurrent

grossPPE: [R],                          // PropertyPlantAndEquipmentGross
accumulatedDepreciation: [R],           // AccumulatedDepreciation...
netPPE: [R],                            // PropertyPlantAndEquipmentNet
goodwill: [R],                          // Goodwill
intangibleAssets: [R],                  // IntangibleAssetsNetExcludingGoodwill
longTermInvestments: [R],               // LongTermInvestments, InvestmentsAndAdvances
deferredTaxAssets: [R],                 // DeferredIncomeTaxAssetsNet
otherNonCurrentAssets: [R],             // OtherAssetsNoncurrent
totalNonCurrentAssets: [D],             // Total Assets – Current Assets

// Liabilities
totalLiabilities: [R],                  // Liabilities
currentLiabilities: [R],                // LiabilitiesCurrent
accountsPayable: [R],                   // AccountsPayableCurrent
accruedLiabilities: [R],                // AccruedLiabilitiesCurrent
shortTermDebt: [R],                     // ShortTermBorrowings, DebtCurrent
currentPortionLTDebt: [R],              // LongTermDebtCurrent
operatingLeaseCurrent: [R],             // OperatingLeaseLiabilityCurrent
financeLeaseCurrent: [R],               // FinanceLeaseLiabilityCurrent
deferredRevenueCurrent: [R],            // DeferredRevenueCurrent
otherCurrentLiabilities: [R],           // OtherLiabilitiesCurrent

longTermDebt: [R],                      // LongTermDebtNoncurrent
operatingLeaseNonCurrent: [R],          // OperatingLeaseLiabilityNoncurrent
financeLeaseNonCurrent: [R],            // FinanceLeaseLiabilityNoncurrent
deferredTaxLiabilities: [R],            // DeferredIncomeTaxLiabilitiesNet
otherNonCurrentLiabilities: [R],        // OtherLiabilitiesNoncurrent
totalNonCurrentLiabilities: [D],        // Total Liab – Current Liab

// Equity
commonStock: [R],                       // CommonStockValue
additionalPaidInCapital: [R],           // AdditionalPaidInCapital
retainedEarnings: [R],                  // RetainedEarningsAccumulatedDeficit
treasuryStock: [R],                     // TreasuryStockValue
aoci: [R],                              // AccumulatedOtherComprehensiveIncomeLossNetOfTax
totalEquity: [R],                       // StockholdersEquity
sharesOutstanding: [R],                 // CommonStockSharesOutstanding (instant)

// Derived
totalDebt: [D],                         // ST Debt + LT Debt + Lease Obligations
netDebt: [D],                           // Total Debt – Cash
},

cashFlow: {
  // Operating
  operatingCashFlow: [R],                // NetCashProvidedByUsedInOperatingActivities
  depreciationAmortization: [R],         // DepreciationDepletionAndAmortization
  stockBasedCompensation: [R],           // ShareBasedCompensation
  deferredIncomeTax: [R],                 // DeferredIncomeTaxExpenseBenefit
  changeInReceivables: [R],              // IncreaseDecreaseInAccountsReceivable

```

```

changeInInventory: [R],           // IncreaseDecreaseInInventories
changeInPayables: [R],           // IncreaseDecreaseInAccountsPayable
changeInOtherWorkingCapital: [D], // Derived from total WC change
otherNonCashItems: [R],          // OtherNoncashIncomeExpense

// Investing
capitalExpenditures: [R],         // PaymentsToAcquirePropertyPlantAndEquipment
investingCashFlow: [R],           // NetCashProvidedByUsedInInvestingActivities
purchaseOfInvestments: [R],       // PaymentsToAcquireInvestments
saleOfInvestments: [R],           // ProceedsFromSaleAndMaturityOfInvestments
purchaseOfBusiness: [R],          // PaymentsToAcquireBusinessesNetOfCashAcquired

// Financing
financingCashFlow: [R],           // NetCashProvidedByUsedInFinancingActivities
dividendsPaid: [R],               // PaymentsOfDividendsCommonStock
shareRepurchases: [R],            // PaymentsForRepurchaseOfCommonStock
debtIssuance: [R],                // ProceedsFromIssuanceOfLongTermDebt
debtRepayment: [R],               // RepaymentsOfLongTermDebt
shortTermDebtNet: [R],            // ProceedsFromRepaymentsOfShortTermDebt
dividendsPerShare: [R],           // CommonStockDividendsPerShareDeclared

// Derived
freeCashFlow: [D],                // Operating CF – CapEx
beginningCash: [R],                // CashCashEquivalentsRestrictedCash... (prior period end)
endingCash: [R],                  // CashCashEquivalentsRestrictedCash...
}
}

```

9.2 Field Count Comparison

Statement	Current Engine	Proposed Schema	Increase
Income Statement	15 fields	23 fields (+4 derived)	+53%
Balance Sheet	19 fields	33 fields (+4 derived)	+74%
Cash Flow	9 fields	22 fields (+2 derived)	+144%
Total	43 fields	78 fields (+10 derived)	+105%

10. Rule One Key Metrics Derivation

Complete Input Requirements

Every Key Metric traces back to financial statement fields. Here's the full dependency chain:

Metrics that need NEW fields (not currently in engine):

Key Metric	Formula	NEW Input Required
Quick Ratio	$(CA - Inventory) / CL$	Already have all
Cash Ratio	$Cash / CL$	Already have all

Current Ratio	CA / CL	Already have all
TIE Ratio	EBIT / Interest Expense	Need EBIT (derived)
Gross Margin	Gross Profit / Revenue	Already have all
EBIT Margin	EBIT / Revenue	Need EBIT
EBITDA Margin	EBITDA / Revenue	Need EBITDA
Operating Profit Margin	Oper Income / Revenue	Already have all
Net Profit Margin	Net Income / Revenue	Already have all
ROE	NI / Avg Equity	Already have all
ROIC	NI / (Equity + LT Debt - Cash)	Already have all
Return on Capital	EBIT / (Equity + Total Debt)	Need EBIT, Total Debt
ROA	NI / Avg Assets	Already have all
Net Debt/Earnings	Net Debt / NI	Need Net Debt
Net Debt/FCF	Net Debt / FCF	Need Net Debt, FCF
Net Debt/Equity	Net Debt / Equity	Need Net Debt
LT Debt/Earnings	LT Debt / NI	Already have all
LT Debt/FCF	LT Debt / FCF	Need FCF
LT Debt/Equity	LT Debt / Equity	Already have all
Debt/Total Capital	Total Debt / (TD + Eq)	Need Total Debt
Asset Turnover	Revenue / Avg Assets	Already have all
Fixed Asset Turnover	Revenue / Avg Net PPE	Already have all
Receivable Turnover	Revenue / Avg Receivables	Need Receivables on BS
Inventory Turnover	COGS / Avg Inventory	Already have all
Payable Turnover	COGS / Avg Payables	Need Accounts Payable
Days metrics	365 / Turnover	From turnover above
Cash Conversion Cycle	DIO + DIR - DIP	From days above
FCF Ratio	FCF / NI	Need FCF
FCF Sales Ratio	FCF / Revenue	Need FCF
BVPS	Equity / Shares Outstanding (EOP)	Need EOP Shares (have it)
OCF per Share	Operating CF / Basic WASO	Already have all
Sales per Share	Revenue / Basic WASO	Already have all

Buybacks per Share	Buybacks / Basic WASO	Already have all
P/E, P/S, P/B, P/CF, P/FCF	Price / (per-share metric)	Need current price (from Polygon)

Summary: Of 61 Key Metrics, the engine already has inputs for ~35. Adding the Priority 1 and Priority 2 fields from Section 7 would cover all 61.

11. Recommendations & Implementation Priority

Phase 1: Critical Additions (enables Rule One Score without manual inputs)

New XBRL tags to add to taxonomy (8 tags):

```
// Balance Sheet additions
{ field: 'accounts_payable', unit: 'USD', tags: ['AccountsPayableCurrent'] }
{ field: 'other_current_assets', unit: 'USD', tags: ['OtherAssetsCurrent',
'PrepaidExpenseAndOtherAssetsCurrent'] }
{ field: 'gross_ppe', unit: 'USD', tags: ['PropertyPlantAndEquipmentGross'] }
{ field: 'accumulated_depreciation', unit: 'USD', tags:
['AccumulatedDepreciationDepletionAndAmortizationPropertyPlantAndEquipment'] }

// Cash Flow additions
{ field: 'deferred_income_tax', unit: 'USD', tags: ['DeferredIncomeTaxExpenseBenefit',
'DeferredIncomeTaxesAndTaxCredits'] }
{ field: 'change_in_receivables', unit: 'USD', tags: ['IncreaseDecreaseInAccountsReceivable',
'IncreaseDecreaseInReceivables'] }
{ field: 'change_in_inventory', unit: 'USD', tags: ['IncreaseDecreaseInInventories'] }
{ field: 'change_in_payables', unit: 'USD', tags: ['IncreaseDecreaseInAccountsPayable',
'IncreaseDecreaseInAccountsPayableAndAccruedLiabilities'] }
```

New derived calculations:

- EBIT = Operating Income + Other Non-Operating Income (or Pre-Tax + Interest Expense)
- EBITDA = EBIT + D&A
- Total Debt = Short-Term Debt + LT Debt + Operating Lease Current + Operating Lease NC + Finance Lease Current + Finance Lease NC
- Net Debt = Total Debt - Cash (including ST Investments)
- Free Cash Flow = Operating CF - CapEx

Phase 2: Full Key Metrics Support (26 additional tags)

```
// Income Statement
{ field: 'interest_income', unit: 'USD', tags: ['InterestIncomeOther', 'InvestmentIncomeInterest',
'InterestAndDividendIncomeOperating'] }
{ field: 'other_income_expense', unit: 'USD', tags: ['OtherNonoperatingIncomeExpense',
'NonoperatingIncomeExpense'] }
{ field: 'depreciation_amortization_is', unit: 'USD', tags: ['DepreciationAndAmortization'] } // income
statement version

// Balance Sheet
{ field: 'accrued_liabilities', unit: 'USD', tags: ['AccruedLiabilitiesCurrent',
```

```

'EmployeeRelatedLiabilitiesCurrent'] }
{ field: 'operating_lease_current', unit: 'USD', tags: ['OperatingLeaseLiabilityCurrent'] }
{ field: 'operating_lease_noncurrent', unit: 'USD', tags: ['OperatingLeaseLiabilityNoncurrent'] }
{ field: 'finance_lease_current', unit: 'USD', tags: ['FinanceLeaseLiabilityCurrent'] }
{ field: 'finance_lease_noncurrent', unit: 'USD', tags: ['FinanceLeaseLiabilityNoncurrent'] }
{ field: 'common_stock_value', unit: 'USD', tags: ['CommonStockValue',
'CommonStocksIncludingAdditionalPaidInCapital'] }
{ field: 'additional_paid_in_capital', unit: 'USD', tags: ['AdditionalPaidInCapital',
'AdditionalPaidInCapitalCommonStock'] }
{ field: 'aoci', unit: 'USD', tags: ['AccumulatedOtherComprehensiveIncomeLossNetOfTax'] }
{ field: 'deferred_tax_liabilities', unit: 'USD', tags: ['DeferredIncomeTaxLiabilitiesNet'] }
{ field: 'deferred_tax_assets', unit: 'USD', tags: ['DeferredIncomeTaxAssetsNet'] }
{ field: 'deferred_revenue_current', unit: 'USD', tags: ['DeferredRevenueCurrent',
'ContractWithCustomerLiabilityCurrent'] }
{ field: 'other_current_liabilities', unit: 'USD', tags: ['OtherLiabilitiesCurrent'] }
{ field: 'other_noncurrent_liabilities', unit: 'USD', tags: ['OtherLiabilitiesNoncurrent'] }
{ field: 'other_noncurrent_assets', unit: 'USD', tags: ['OtherAssetsNoncurrent'] }
{ field: 'preferred_stock', unit: 'USD', tags: ['PreferredStockValue'] }

// Cash Flow
{ field: 'purchase_of_investments', unit: 'USD', tags: ['PaymentsToAcquireInvestments',
'PaymentsToAcquireAvailableForSaleSecuritiesDebt'] }
{ field: 'sale_of_investments', unit: 'USD', tags: ['ProceedsFromSaleAndMaturityOfInvestments',
'ProceedsFromSaleOfAvailableForSaleSecuritiesDebt'] }
{ field: 'purchase_of_business', unit: 'USD', tags: ['PaymentsToAcquireBusinessesNetOfCashAcquired',
'PaymentsToAcquireBusinessesGross'] }
{ field: 'debt_issuance', unit: 'USD', tags: ['ProceedsFromIssuanceOfLongTermDebt',
'ProceedsFromIssuanceOfDebt'] }
{ field: 'debt_repayment', unit: 'USD', tags: ['RepaymentsOfLongTermDebt', 'RepaymentsOfDebt'] }
{ field: 'short_term_debt_net', unit: 'USD', tags: ['ProceedsFromRepaymentsOfShortTermDebt',
'ProceedsFromRepaymentsOfCommercialPaper'] }
{ field: 'other_noncash_items', unit: 'USD', tags: ['OtherNoncashIncomeExpense'] }

```

Phase 3: TTM Computation

TTM (Trailing Twelve Months) requires quarterly data. The approach:

1. Fetch quarterly facts from EDGAR (same API, filter for 10-Q filings with `form: "10-Q"` and quarterly periods)
2. Sum the 4 most recent quarters for income/cash flow items
3. Use the most recent quarter-end for balance sheet items
4. Handle fiscal year boundaries correctly

Rule One Toolbox includes TTM as the first column in consolidated financials. This is critical for current-period analysis.

Phase 4: Move Dividend Per Share from Cash Flow to Income Statement

Currently `dividends_per_share` is in `CASHFLOW_TAXONOMY`. It should also appear in `INCOME_TAXONOMY` since Rule One Toolbox shows it on the income statement and it's needed for per-share metrics. The same XBRL tag (`CommonStockDividendsPerShareDeclared`) works for both.

Implementation Notes

1. **Backward compatibility:** The expanded schema is additive — all existing field names remain unchanged. No breaking changes to `growthRates.js`, `returnMetrics.js`, `ruleOneScore.js`, or `valuation.js`.

2. **Eliminating manual inputs:** With the expanded taxonomy, these fields that currently require `manualInputs` can be auto-populated:
- `manualInputs.cash` → `balanceSheet.cashAndEquivalents` or `cashAndShortTermInvestments`
 - `manualInputs.capEx` → `cashFlow.capitalExpenditures`
 - `manualInputs.retainedEarnings` → `balanceSheet.retainedEarnings`
 - FCF is derived automatically
3. **Tag fallback strategy:** Continue using the merge strategy (first tag wins per year, later tags fill gaps). This handles ASC 606 revenue transitions and similar standard changes seamlessly.
4. **Unit handling:** All new balance sheet tags are `instant/USD` . All new cash flow tags are `duration/USD` . The existing `extractAnnualFact` function handles both correctly.
-

Appendix A: Complete XBRL Tag Reference

Tags Currently in Engine (36 total)

Income (15): RevenueFromContractWithCustomerExcludingAssessedTax, Revenues, SalesRevenueNet, SalesRevenueGoodsNet, RevenueFromContractWithCustomerIncludingAssessedTax, CostOfRevenue, CostOfGoodsAndServicesSold, CostOfGoodsSold, GrossProfit, SellingGeneralAndAdministrativeExpense, SellingAndMarketingExpense, GeneralAndAdministrativeExpense, ResearchAndDevelopmentExpense, ResearchAndDevelopmentExpenseExcludingAcquiredInProcessCost, OperatingExpenses, CostsAndExpenses, OperatingIncomeLoss, InterestExpense, InterestExpenseDebt, InterestIncomeExpenseNonoperatingNet, InterestIncomeExpenseNet, IncomeLossFromContinuingOperationsBeforeIncomeTaxesExtraordinaryItemsNoncontrollingInterest, IncomeLossFromContinuingOperationsBeforeIncomeTaxesMinorityInterestAndIncomeLossFromEquityMethodInvestments, IncomeLossFromContinuingOperationsBeforeIncomeTaxesDomestic, IncomeTaxExpenseBenefit, NetIncomeLoss, ProfitLoss, NetIncomeLossAvailableToCommonStockholdersBasic, EarningsPerShareDiluted, EarningsPerShareBasic, WeightedAverageNumberOfDilutedSharesOutstanding, WeightedAverageNumberOfShareOutstandingBasicAndDiluted, WeightedAverageNumberOfSharesOutstandingBasic

Balance (19): Assets, AssetsCurrent, CashAndCashEquivalentsAtCarryingValue, CashCashEquivalentsAndShortTermInvestments, Cash, ShortTermInvestments, MarketableSecuritiesCurrent, AvailableForSaleSecuritiesCurrent, AccountsReceivableNetCurrent, ReceivablesNetCurrent, AccountsReceivableNet, InventoryNet, InventoryFinishedGoodsAndWorkInProcess, InventoryRawMaterialsAndSupplies, PropertyPlantAndEquipmentNet, Goodwill, IntangibleAssetsNetExcludingGoodwill, FiniteLivedIntangibleAssetsNet, LongTermInvestments, InvestmentsAndAdvances, MarketableSecuritiesNoncurrent, Liabilities, LiabilitiesCurrent, ShortTermBorrowings, DebtCurrent, LongTermDebtCurrent, LongTermDebtNoncurrent, LongTermDebt, LongTermDebtAndCapitalLeaseObligations, StockholdersEquity, StockholdersEquityIncludingPortionAttributableToNoncontrollingInterest, RetainedEarningsAccumulatedDeficit, RetainedEarningsUnappropriated, TreasuryStockValue, TreasuryStockCommonValue, CommonStockSharesOutstanding, CommonStockSharesIssued

Cash Flow (13): NetCashProvidedByUsedInOperatingActivities, NetCashProvidedByUsedInOperatingActivitiesContinuingOperations, PaymentsToAcquirePropertyPlantAndEquipment, PaymentsToAcquireProductiveAssets, DepreciationDepletionAndAmortization, DepreciationAndAmortization, DepreciationAmortizationAndAccretionNet, Depreciation, ShareBasedCompensation, AllocatedShareBasedCompensationExpense, NetCashProvidedByUsedInInvestingActivities, NetCashProvidedByUsedInInvestingActivitiesContinuingOperations, NetCashProvidedByUsedInFinancingActivities, NetCashProvidedByUsedInFinancingActivitiesContinuingOperations, PaymentsOfDividendsCommonStock, PaymentsOfDividends, PaymentsOfOrdinaryDividends, CommonStockDividendsPerShareDeclared, CommonStockDividendsPerShareCashPaid, PaymentsForRepurchaseOfCommonStock, PaymentsForRepurchaseOfEquity

Tags to Add (34 new unique tags)

Income (3): InterestIncomeOther, InvestmentIncomeInterest, OtherNonoperatingIncomeExpense, NonoperatingIncomeExpense

Balance (18): OtherAssetsCurrent, PrepaidExpenseAndOtherAssetsCurrent, PropertyPlantAndEquipmentGross, AccumulatedDepreciationDepletionAndAmortizationPropertyPlantAndEquipment, AccountsPayableCurrent, AccruedLiabilitiesCurrent, OperatingLeaseLiabilityCurrent, OperatingLeaseLiabilityNoncurrent, FinanceLeaseLiabilityCurrent, FinanceLeaseLiabilityNoncurrent, DeferredIncomeTaxLiabilitiesNet, DeferredIncomeTaxAssetsNet, DeferredRevenueCurrent, ContractWithCustomerLiabilityCurrent, OtherLiabilitiesCurrent, OtherLiabilitiesNoncurrent, OtherAssetsNoncurrent, CommonStockValue, AdditionalPaidInCapital, AdditionalPaidInCapitalCommonStock, AccumulatedOtherComprehensiveIncomeLossNetOfTax, PreferredStockValue

Cash Flow (13): DeferredIncomeTaxExpenseBenefit, IncreaseDecreaseInAccountsReceivable, IncreaseDecreaseInReceivables, IncreaseDecreaseInInventories, IncreaseDecreaseInAccountsPayable, IncreaseDecreaseInAccountsPayableAndAccruedLiabilities, PaymentsToAcquireInvestments, ProceedsFromSaleAndMaturityOfInvestments, PaymentsToAcquireBusinessesNetOfCashAcquired, ProceedsFromIssuanceOfLongTermDebt, RepaymentsOfLongTermDebt, ProceedsFromRepaymentsOfShortTermDebt, OtherNoncashIncomeExpense

Appendix B: XBRL Taxonomy Reference Tools

These resources are useful for looking up XBRL tag definitions and verifying tag availability:

Resource	URL	Notes
Calcbench Element Browser	calcbench.com/element/{ElementName}	Definitions, period/balance type, parent/child hierarchy for any us-gaap element
FASB Taxonomy Viewer (Yeti)	xbrlview.fasb.org/yeti/	Official FASB tool to browse the complete taxonomy tree
XBRL US Taxonomy Pages	xbrl.us/xbrl-taxonomy/2025-us-gaap/	Links to viewers, schema files, documentation
SEC EDGAR XBRL Guide	sec.gov/files/edgar/filer-information/specifications/xbrl-guide.pdf	Official SEC implementation guide
DQC Element Lists	xbrl.us/data-rule/dqc_0015-le/	Data Quality Committee valid element lists per statement
FASB Taxonomy Schema	xbrl.fasb.org/us-gaap/2024/elts/us-gaap-2024.xsd	Raw XSD schema with all ~17,000 element definitions

Appendix C: Working Capital Change Tag Reference

Working capital change tags on the cash flow statement follow specific conventions:

Tag	Scope	Balance	Notes
IncreaseDecreaseInAccountsReceivable	Trade receivables only	Credit	More specific
IncreaseDecreaseInReceivables	All receivables	Credit	Parent — use as fallback
IncreaseDecreaseInInventories	All inventory	Credit	Standard, widely used
IncreaseDecreaseInAccountsPayable	Trade payables only	Debit	More specific

IncreaseDecreaseInAccountsPayableAndAccruedLiabilities	Payables + accruals	Debit	Parent — many companies use this combined version
IncreaseDecreaseInOtherOperatingCapitalNet	Catch-all "other"	Credit	Other working capital adjustments
IncreaseDecreaseInPrepaidDeferredExpenseAndOtherAssets	Prepays + other assets	Credit	Common additional WC line
IncreaseDecreaseInAccruedLiabilities	Accrued liabilities only	Debit	Separate from accounts payable

Sign convention for IncreaseDecrease tags: Positive value = increase in the item. For assets (AR, inventory), an increase uses cash (negative operating impact). For liabilities (AP, accruals), an increase provides cash (positive operating impact). EDGAR reports these as-is — the sign already reflects the cash flow impact correctly.

Appendix D: Equity Tag Reference

Tag	What It Represents	Notes
CommonStockValue	Par value only (often \$0.001 × shares = tiny number)	NOT total equity
CommonStocksIncludingAdditionalPaidInCapital	Par + APIC combined	Some companies (AAPL) use this single tag
AdditionalPaidInCapital	APIC for all stock classes	More common
AdditionalPaidInCapitalCommonStock	APIC for common stock only	More specific; SFM uses this
TreasuryStockValue	Dollar amount of repurchased shares	Debit balance — reduces equity
TreasuryStockCommonShares	Share count in treasury	Replaces deprecated TreasuryStockShares
StockRepurchasedAndRetiredDuringPeriodShares	Shares bought back and retired (period)	Duration; for companies that retire shares
TreasuryStockSharesAcquired	Shares bought back and held as treasury (period)	Duration; for companies that hold treasury stock
AccumulatedOtherComprehensiveIncomeLossNetOfTax	AOCI — unrealized gains/losses, FX translation, pension	Credit balance

End of Report